Olist E-Commerce Analytics

Sales, Delivery & Customer Satisfaction

DEPI Professional Data Analytics Capstone

Python | SQL Server | Power BI

Karim - Asmaa - Ahmed Anany - Mohamed Ali

Problem & Objectives

- Olist: Brazilian marketplace, ~100K orders (Sep 2016 - Oct 2018)
- Raw data: 9 CSV files with quality issues
- Goal: turn it into decisions for the business
  - How do sales evolve and what drives them?
  - How reliable is delivery?
  - Does delivery affect satisfaction?
  - Do customers come back?

Team & Workflow

Karim	SQL Server: database & star schema, KPI views
Asmaa	Python: cleaning, EDA, cohort retention
Ahmed Anany	Power BI: dashboard & DAX measures
Mohamed Ali	Analytical questions, RFM, final report

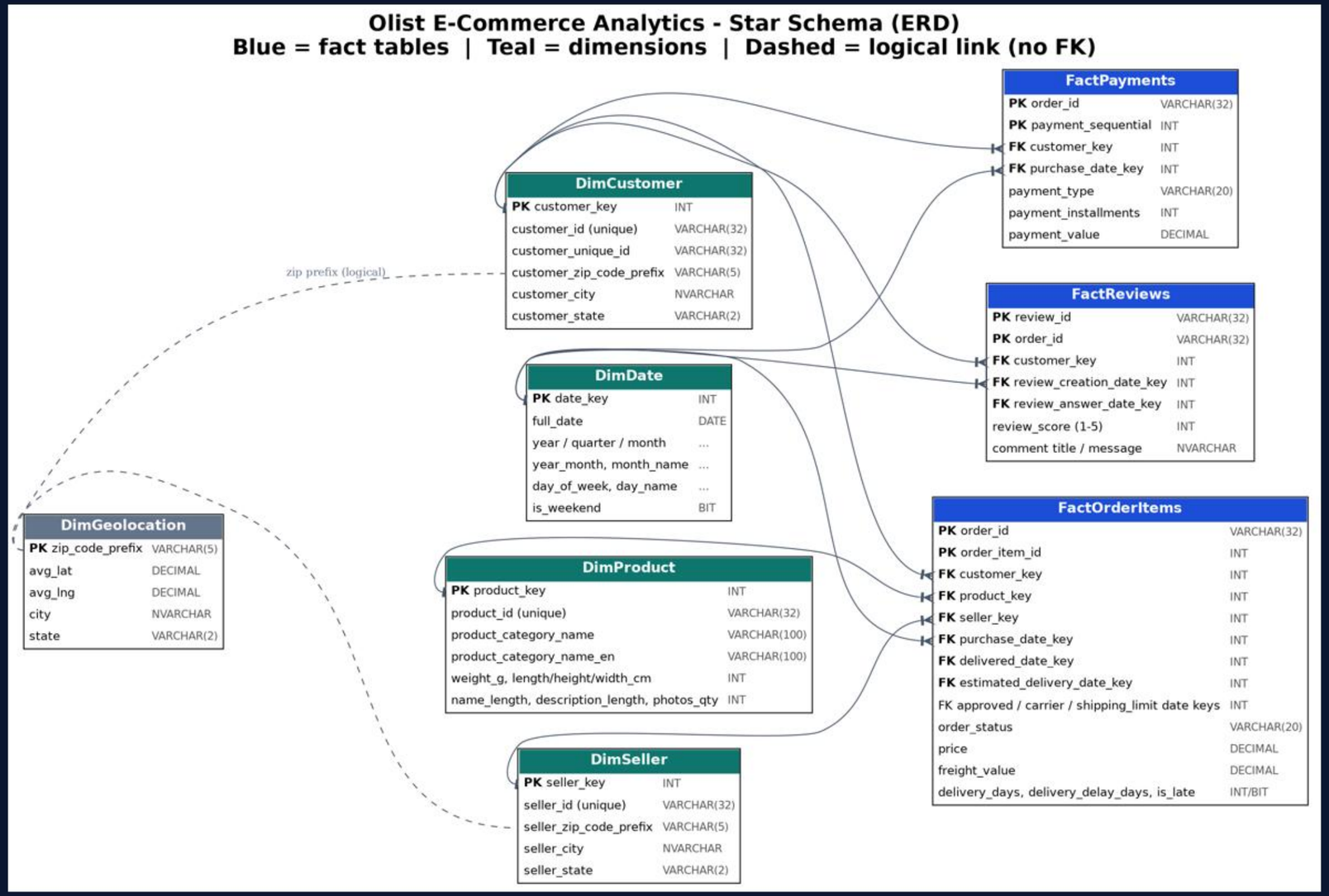
28-day plan | shared Drive folders | weekly deliveries between members

Data & Cleaning (Python)

99,441	9	2016-2018
orders	CSV files	period

- Dates stored as text -> converted to datetime
- Nulls: 2,965 undelivered, 610 products w/o category
- 814 duplicate review_ids, 551 orders with 2+ reviews
- Category names translated to English; typos fixed
- Canceled orders excluded from KPIs

Database Design: Star Schema (SQL Server)



Key KPIs

$13.50M
Revenue (excl. canceled)

98,205
Valid orders

$137.4
Avg order value

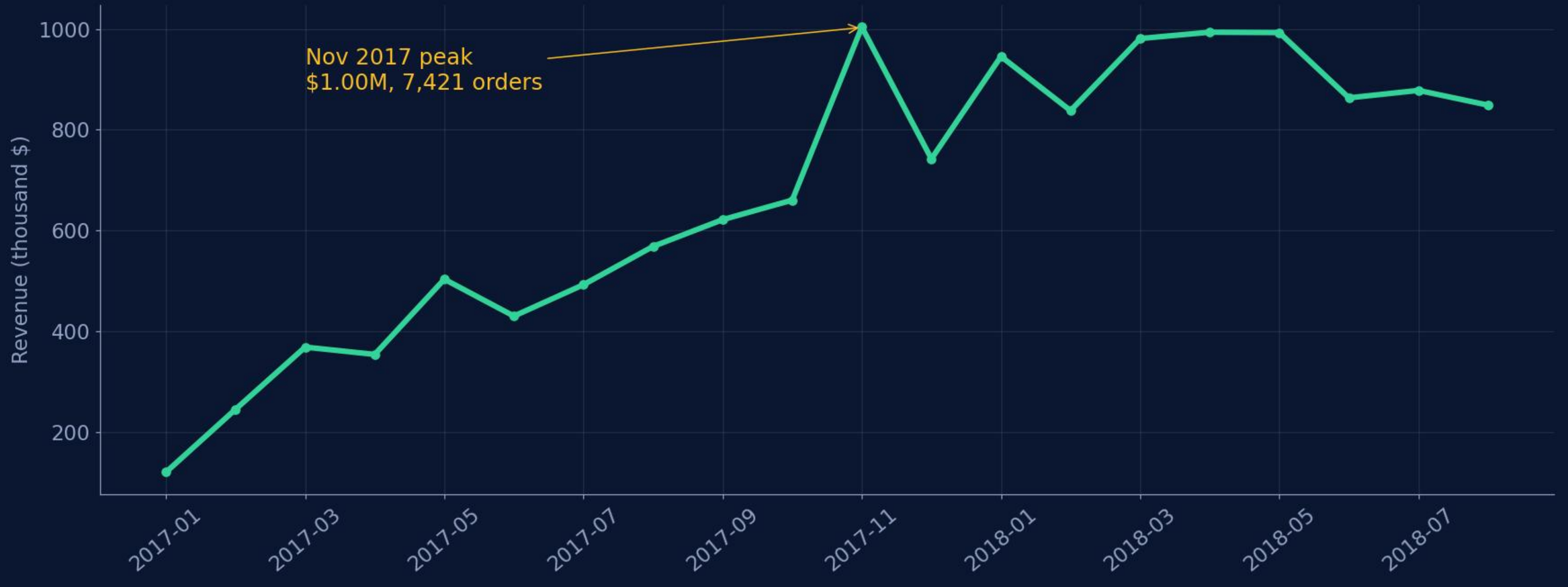
12.1 days
Avg delivery time

4.09 / 5
Avg review score

8.1%
Late deliveries

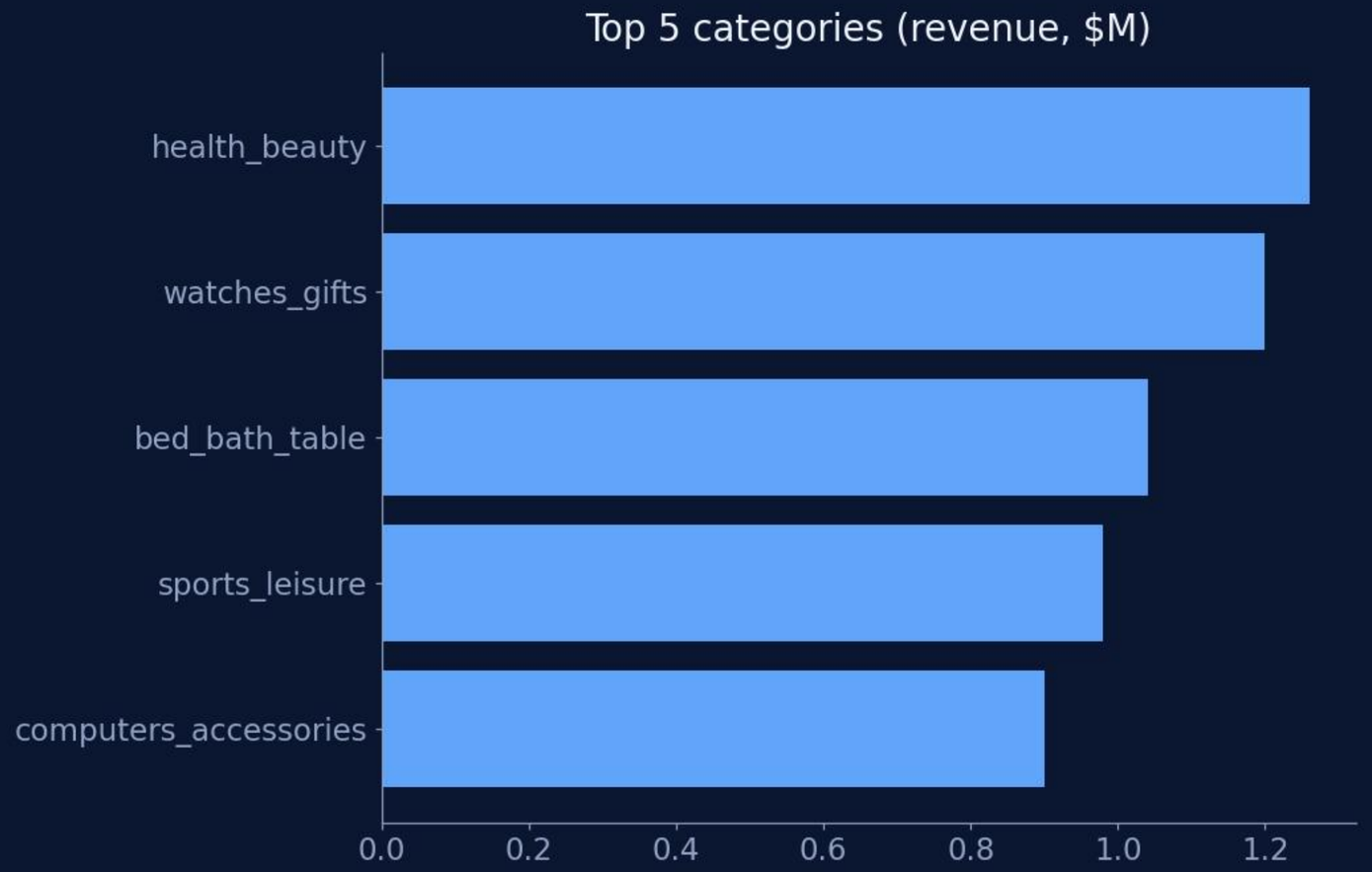
Revenue = sum of item price (freight excluded) | Valid orders = status not 'canceled'

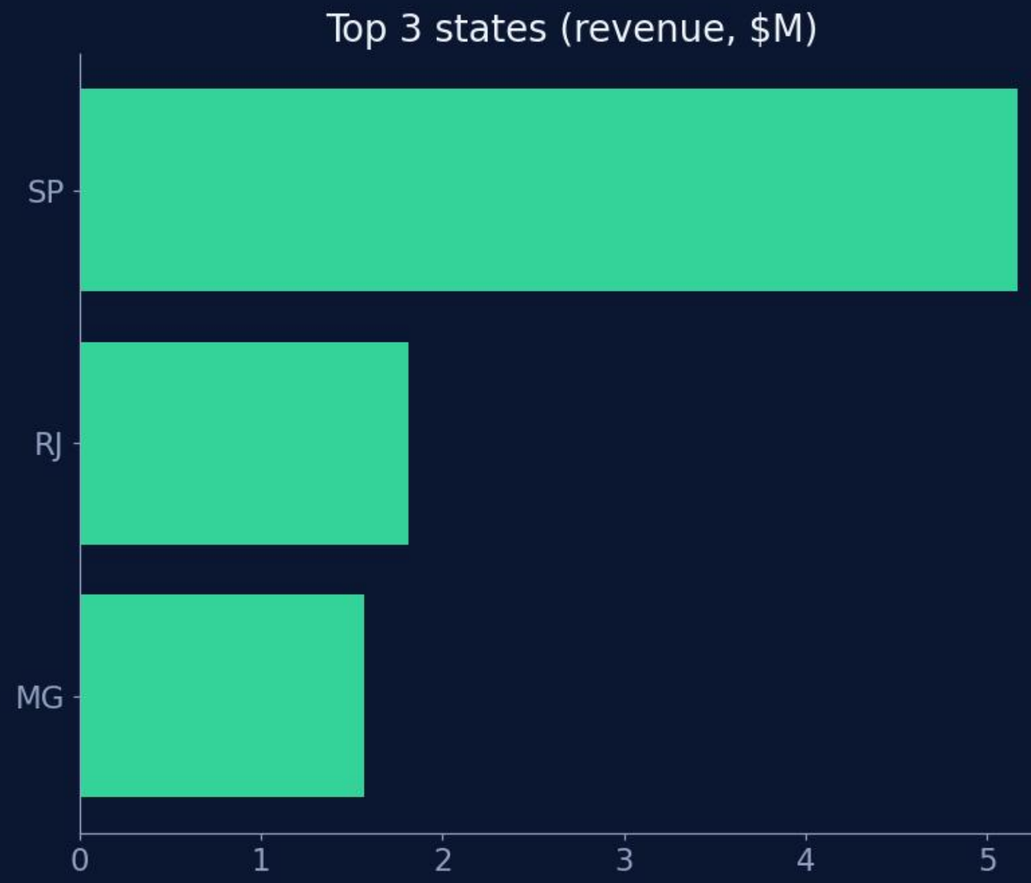
Sales Trend



Revenue grew from $6.1M (2017) to $7.3M (2018, partial data to Aug). Peak in Nov 2017 (likely Black Friday).

Products & Geography





Top 5 categories = 40% of revenue. Sao Paulo alone = 38%; SP + RJ + MG = 63%. Credit card = 78% of payment value.

Delivery & Customer Satisfaction



- 96,470 delivered orders
- Avg delivery: 12.1 days
- 8.1% arrive after the estimate
- Late orders score 2.57 vs 4.29 on time
- Worst late rates: AL 24%, MA 20%, PI 16%

Customer Retention

96,096
unique customers

3.1%
placed 2+ orders

14.7%
reviews scored 1-2

- Almost all customers buy once: growth depends on new customers
- Cohort retention (Python) and RFM segmentation show where repeat buyers are
- Customer = customer_unique_id (not customer_id)

Insights & Recommendations

Fix delivery in slow states

Late orders drop review scores by ~1.7 points; target AL, MA, PI first.

Build repeat purchase

Only 3% re-buy: launch post-purchase offers and loyalty for top RFM segments.

Plan around Nov peak

Stock and logistics capacity for the Black Friday surge.

Reduce concentration

SP = 38% of revenue: grow RJ, MG and other regions.

Thank You

github.com/DepiProject2/Ecommerce-analytics

Questions?